MILITARY SPECIFICATION

WEBBING, TEXTILE, WOVEN NYLON, ROLLED SELVAGE

1. SCOPE

1.1 Scope. This specification covers untreated, undyed, nylon webbing.

1.2 Classification. The nylon webbing shall be of the following types, as specified (see 3.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-43334	Webbing and Tape, Textile, Packaging and Packing of

STANDARDS

Federal

Fed. Test Method Std. 191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Material

3.1.1 Nylon yarn. The nylon yarn used in the manufacture of all types of webbing shall be a bright, high tenacity, light and heat resistant, polyamide prepared from hexamethylene diamine and adipic acid, or its derivatives. It shall have a minimum melting point of 472°F. The yarn for the selvages shall be preshrunk prior to weaving. The yarn shall not be bleached in any manner or in any subsequent process.

3.2 Physical properties. The construction and physical properties of the finished webbing shall be as specified in table I. The supplier shall certify that the webbing or yarn, separately, has not been subjected to any type of bleaching.

Table I. Construction and physical properties

Characteristic	Type I	Type II	Type III	Type IV	Type V
Width (including selvages); inches	2+0-1/16	2+0-1/16	2+0-1/16	2+0-1/16	2+0-1/16
Selvage width, inches 1/ Maximum	1/8	1/8	5/32	7/32	1/8
Ends (Minimum)	302	152	208	280	448
Ends in each selvage	40	20	28	32	56
Picks/inch (Minimum)	36	36	36	24	36
Weight, ounces/yd. (Maximum)	0.43	0.82	1.20	1.40	0.56

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Table I (continued)

Characteristic	Type I	Type II	Type III	Type IV	Type V
Yarn denier and ply					
Warp	210	840	840	840	210
Filling	420	420	420	840	420
Selvage	210/2	840/2	840/2	840/2	210/2
Twist, turns/inch (Minimum)					
Warp	2.5	2.5	2.5	2.5	2.5
Filling	2.5	2.5	2.5	2.5	2.5
Selvage	2.5	2.5	2.5	2.5	2.5
Breaking strength (Minimum) 2/	1000	2200	3000	3500	1500

1/ This requirement shall apply to each selvage edge, separately.

2/ No single determination shall be less than the minimum.

3.3 Weave. The weave for all types of webbing shall be as shown in figure 1.

3.4 Nonfibrous material. The water-soluble and chloroform-soluble materials shall be no more than 2.0 percent.

3.5 Length of roll. Unless otherwise specified, each roll shall contain 100 ± 10 yards. A roll shall contain no more than 2 pieces and each piece shall be not less than 10 yards in length (see 6.2).

3.6 Identification tickets. Each roll shall have a ticket attached with not finer than 5-ply cotton string doubled to not less than 6 inches long. The tickets shall be made of paperboard not less than 0.015 inch in thickness and the color shall be manila or light in intensity to permit easy reading of printed, stamped or typed markings. The use of handwritten entries is prohibited. The ticket shall have clipped corners at the end where a reinforcing patch (with or without a metal eyelet) is firmly affixed for attaching the tying string. The ticket shall be legibly printed with water insoluble ink with the following information:

Stock number
Nomenclature
Specification number
Yardage
Contract number and date of manufacture (month and year)

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SELVAGE WEAVE - TWO ENDS WEAVING AS ONE

FIGURE 1. WEAVE DIAGRAM

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Supplier's name
Name of contracting agency

3.7 Fiber identification. Each roll of webbing shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.8 Workmanship. The finished webbing shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. The quality conformance inspection shall consist of the examinations and tests specified herein. Except where otherwise specified, inspection shall be in accordance with MIL-STD-105.

4.2.1 Component inspection. The testing of components shall be as specified in table II. The lot size of the webbing and the number of component samples per lot size shall be as specified herein. The lot shall be unacceptable if one or more samples fail to meet any requirements of this specification.

Lot size (yards)	Number of samples
800 or less	2
801 up to and including 22,000	3
22,001 - 50,000	4
50,001 - 75,000	5
75,001 - 100,000	6
100,000 and over	6 + 1 sample for each 25,000 yard increment

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Table II. Testing of components

Characteristic	Requirement reference	Test Method FTMS 191
Yarn		
Warp, filling and selvage:		
Nylon identification	3.1.1	1/
Heat resistance	3.1.1	1/
Light resistance	3.1.1	1/
Luster	3.1.1	1/
Melting point	3.1.1	1534
Preshrinkage	3.1.1	1/
Tenacity	3.1.1	1/
Unbleached	3.1.1	1/
Denier	Table I	4021
Ply	Table I	1/
Twist (turns/inch)	Table I	4054

1/ A certificate of compliance shall be submitted for these requirements and it shall include: A certified statement that the yarn used is that specified in 3.1.1 along with the yarn producer's type, identification data and melting point and supported by a certified copy of the yarn producer's certification to the webbing manufacturer.

4.2.2 Examination of the end item. Defects found during examinations shall be classified in accordance with 4.2.2.1 through 4.2.2.2.2.

4.2.2.1 Yard-by-yard examination. The webbing shall be examined on both sides and visual defects classified as listed in table III. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the webbing, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be one linear yard. The sample size shall be in accordance with level III of MIL-STD-105. The acceptable quality level shall be 2.5 for minor defects and 0.65 critical defects per 100 units (yards). An approximate equal number of yards shall be examined from each roll selected. The number of rolls from which the sample is to be selected shall be in accordance with table IV.

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Table III. Classification of defects

Defect	Description	Critical	Minor
Abrasion marks	Resulting in rupture of yarns, or in nap sufficient to obscure the identity of any yarn exceeding 10 percent of width or 1 inch in length.	X	
Broken or missing	Two or more regardless of length or a single and exceeding 6 inches in length.	X	
	Single end under 6 inches but exceeding ½ inch.		X
Broken or missing end	Two or more regardless of extent.	X	
NOTE:	The filling tie-in or joining shall not be construed as a defect of any nature.		
Coarse or light filling bar	Resulting in visible difference in stiffness or thickness of webbing and extending for more than ½ inch in the length direction.	X	
	Resulting in visible difference in stiffness or thickness of webbing and extending for ½ inch or less in the length direction.		X
Twist or distortion	Webbing will not lay flat upon application of manual pressure due to twist or distortion.		X
Cut, hole, or tear	Any cut, hole or tear	X	
Drop-ply	Clearly visible on more than 2 ends within same length and extending over 9 linear inches or more. 1/	X	
	Clearly visible on 1 or 2 ends within same length and extending over 9 linear inches or more. 1/		X

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Table III. (continued)

Defect	Description	Critical	Minor
Edges	Frayed, slack or otherwise poorly constructed and exceeding ¼ inch in length.	X	
Floats or skips	Three or more, ½ inch or more in combined warp and filling directions or single float or skip over more than 1 inch.	X	
	Three or more, less than ½ inch in combined warp or filling directions or single float or skip over more than ½ inch but not exceeding 1 inch, if in warp, or more than ¼ inch of width but not exceeding 1 inch if in filling.		X
Hitchback crack	Clearly visible opening between adjoining picks, or warpwise tension area over part of the width resulting in visible light and heave places. 1/		X
Jerked-in filling, slough-off, slug	A clearly visible loop of filling pulled in at edges. 1/		X
Kinks	More than 3 in any 9 linear inches.	X	
Knots	More than 1 knot in any 9 linear inches.	X	
	One knot every 2 yards with untrimmed ends extending from surface of webbing.		X
Mispick, double pick	Two or more across the full width.	X	
	Single across the full width.		X
Slack end	Two or more in the same length, jerked in between picks, or forming clearly visible loops on the surface.	X	
	Single jerked in between picks or forming clearly visible loops on the surface.		X

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Table III. (continued)

Defect	Description	Critical	Minor
Slub, or slug, gout	More than twice the thickness of the yarn (or ply, if plied)		X
Smash	Any smash.	X	
Spot, stain or streak	Any clearly visible. 1/		X
Tight end	A minimum of 12 inches in length. 1/		X
Uneven weaving through out	Two or more ends or picks extending for more than 12 inches in length. 1/		
Wrong draw	Extending for more than 9 inches.	X	

1/ Clearly visible at normal inspection distance (approximately 3 feet).

Table IV. Sampling plan

Lot size in yards	Sample size in rolls	Maximum number defects acceptable in sample (see 4.2.2.2)
up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to & including 10,000	8	0
10,001 up to & including 35,000	13	0
35,001 up to & including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than 3 rolls, each roll in the lot shall be examined.

4.2.2.2 Length examination

4.2.2.2.1 Examination for length of individual roll. Each roll in the sample shall be examined for the defects listed below. The sample unit for this examination shall be one roll. The sample size and acceptance shall be as shown in table V.

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Defects

Gross length less than specified minimum length or more than specified maximum length.

Gross length more than 2 yards less than gross length marked on piece ticket.

Any piece less than 10 yards in length.

Any roll containing more than 3 pieces.

4.2.2.2.2 Examination for total yardage in sample. The lot shall be unacceptable if the total of the actual gross lengths of rolls in the samples selected in accordance with table IV is less than the total of the gross lengths marked on roll tickets.

4.3 Examination for compliance with Textile Fiber Products Identification Act. Each roll of the webbing shall be examined for conformance to the Textile Fiber Products Identification Act.

4.4 Examination of preparation for delivery. Preparation for delivery shall be inspected for conformance to section 5.

4.5 Testing of the end item. The methods of testing specified in FTMS 191, wherever applicable, and as listed in table V shall be followed. Except for breaking strength the physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values used in obtaining the final results. The sample unit for testing shall be 8 yards. The lot size and the number of samples per lot size shall be as follows:

Lot size (yards)	Number of samples
800 or less	2
801 up to and including 22,000	3
22,001 and over	3 + 1 each additional sample from each additional successive increment of 50,000 yards produced for the lot.

The lot shall be unacceptable if one or more samples fail to meet any requirements of this specification.

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Table V. Test methods

Characteristics	Requirement reference	Test Method FTMS 191	Number of determinations per sample unit	Results reported as
Weight	Table I	5041	-	To nearest 0.01 ounce
Yarn				
Ends, body	Table I	5050	-	To nearest whole number
Ends, selvage	Table I	5050	-	To nearest whole number
Picks per inch	Table I	5050	-	To nearest whole number
Unbleached	3.2	1/	-	Pass or fail
Breaking strength	Table I	2/ 5100	-	Individual results for each specimen tested to nearest 1.0 pound
Weave	3.3	Visual	1	Pass or fail
Nonfibrous material	3.4	2611	-	To nearest 0.1 percent
Width				
Overall	3.2	4.6	5	To nearest 1/32 inch
Selvage	3.2	4.6	5	To nearest 1/32 inch

1/ The supplier's certificate of compliance will be accepted for this characteristic.

2/ Breaking strength shall be made in accordance with method 5100 except that the width of the jaws shall not be less than the width of the specimen. Split drum type of jaws shall be used and the no-load rate of jaw separation shall be 2 to 4 inches per minute. The distance between jaws at the start of the test shall be 10 $\pm$ ½ inches. The minimum length of the specimen for tests shall be 45 inches.

4.6 Measurement of width. The sample size for this determination shall be five (5) yards. One measurement shall be made in each one (1) yard length

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increment of the sample. The material shall be placed flat, with hand tensioning, on a smooth horizontal surface. For the overall width the distance from edge to edge shall be measured to the nearest 1/32 inch in a line perpendicular to the lengthwise direction of the material. For the selvage width the distance from the outside edge to the inside edge of the selvage shall be measured to the nearest 1/32 inch in a line perpendicular to the lengthwise direction of the material. Determinations shall be made on each selvage edge and reported separately. The width of the sample shall be the average of the five measurements.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.1.1 Levels A, B and C. Webbing, put up as specified shall be packaged in accordance with the applicable requirements of MIL-W-43334.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2 and 6.2.1).

5.2.1 Levels A, B and C. Webbing shall be packed in accordance with the applicable requirement of MIL-W-43334.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-W-43334.

6. NOTES

6.1 Intended use. The webbing covered by this specification is intended for use in heavy duty high-speed ribbon parachute canopies.

6.2 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification.

b. Type.

c. Length of roll required (if other than specified in 3.5).

d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives, as indicated in MIL-W-43334, may be exercised in the preparation for delivery of webbing.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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